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PROJECT REPORT

ON

FOOD DELIVERY TIME PREDICTION

An End-to-End Predictive Analytics / KDD Project

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Submitted To

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Course

Predictive Analytics

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1. Introduction

Problem statement

Customers and delivery operators benefit from a realistic estimate of how long an order will take. This project predicts total delivery duration in minutes from distance, traffic, weather, time of day, vehicle type and courier experience. The target is numerical, so the problem is supervised regression rather than classification. The intended output is an initial estimate that supports communication and planning.

Objectives

The project demonstrates the complete Knowledge Discovery in Databases (KDD) process: problem definition, data selection, understanding and cleaning, transformation, model development, evaluation, interpretation and reuse. It compares a simple baseline with several predictive models, handles missing inputs within training pipelines and produces an executable notebook and a saved estimator.

Prediction boundary and scope

Recorded preparation duration is deliberately excluded. The source does not establish that this value was available when an order was placed; using a completed preparation time could give an early estimate information from the future. Order ID is excluded because it identifies a record rather than describing the delivery process. The remaining six predictors are treated as supplied contextual information.

Weather and traffic must also be available at the intended prediction time in a live application. The present benchmark does not verify that availability. It therefore measures performance under the supplied conditions, not the accuracy of a fully deployed delivery service. No timestamps or entity identifiers are available for temporal or restaurant-level testing.

Success criterion and report structure

The model is selected using the lowest mean five-fold validation MAE on the development subset. A separate holdout supplies the final MAE, RMSE, R-squared and proportion of predictions within ten minutes. The holdout is not used to choose the configuration. Improvements are measured against a training-median baseline.

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2. Literature Review

Food-delivery regression research

Garg, Ayaan, Parekh and Udandarao (2025) compare linear and ensemble approaches for food-delivery time estimation in an Indian urban context. Their public abstract highlights contextual and geographic information and discusses models including random forests and boosting. This supports testing both an additive model and nonlinear alternatives. The present experiment uses a different public dataset, feature boundary and split, so its results are not directly comparable with that paper. [2]

Route and time prediction

Wen and colleagues (2023) survey instant-delivery route and arrival-time prediction. Their taxonomy distinguishes route-only, time-only and joint tasks and considers different architectures and learning paradigms. This project addresses only a time-regression task: it does not optimize courier routes or assign orders. The survey helps define that narrower scope. [3]

Preprocessing, evaluation and explanation

The scikit-learn pipeline documentation explains how transformations and estimators can be combined in a single fitted object. This design helps keep preprocessing statistics within the training partition. Cross-validation provides a model-selection mechanism, while a held-out set offers a separate final evaluation. This project applies those principles with fold-local imputation and encoding. [4,5]

Permutation importance measures a fitted model's reliance on a feature by shuffling it and examining score deterioration. It is useful for interpretation, but it does not measure causal effects. Correlation and unrealistic combinations after shuffling must be considered when reading the ranking. [6]

Connection to this implementation

Source	Application	Limit
Garg et al.	Compare linear and ensemble regression	Different data; no benchmark scores copied
Wen et al.	Define time prediction separately from routing	No route or dispatch optimization
Pipeline / CV guides	Fit transformations inside each fold	Random split assumes comparable independent orders
Importance guide	Inspect model dependence on raw inputs	Associations are not causes

The contribution is a transparent educational workflow rather than a new algorithm. The six-input design gives a clear prediction boundary, and each reported result can be traced to saved notebook outputs. The literature discussion uses the cited public abstracts and official documentation.

3. Workflow / Predictive Modelling Process

The dataset is Food Delivery Time Prediction by Den_Kuznetz on Kaggle, version 1. The provider metadata lists Apache 2.0. The source describes the columns but does not document a collection period or sampling protocol, so the project treats it as an educational benchmark rather than verified company data. [1]

The audit found **1,000 orders and 9 columns**, with 120 missing cells: 30 each in weather, traffic, time of day and courier experience. Targets are complete. No duplicate order IDs or repeated modelling records were found. Numeric ranges were checked; valid long-duration records were retained.

Stage	Implementation
Select and split	Six inputs; exclude ID and preparation duration. Fixed 80/20 split: 800 development and 200 test orders, seed 42.
Clean and transform	Median numeric imputation plus missing indicators and scaling; categorical missing values become Unknown, followed by one-hot encoding.
Develop models	Five-fold shuffled cross-validation within development only; preprocessing is fitted separately in each fold.
Choose and evaluate	Select lowest mean validation MAE; refit on all development rows; score the untouched holdout once.
Interpret and reuse	Residual analysis, traffic-group errors and permutation importance; export the complete pipeline and verify reload predictions.

Predetermined model configurations

Family	Settings
Median baseline	Always predicts the training-target median.
Ridge regression	Regularization alpha: 1 or 10.
Random forest	200 trees; minimum leaf size: 2 or 5; seed 42.
Histogram gradient boosting	200 iterations; learning rate 0.05; maximum leaves: 7 or 15; early stopping disabled; seed 42.

All candidates use identical partitions. Negative outputs are clipped to zero consistently, without rounding. The five-fold standard deviation describes score variation, not a confidence interval. Without timestamps or entity IDs, the random split cannot demonstrate generalization to a later month, a new courier or a new city.

4. Results and Discussion

4.1 Candidate comparison and holdout performance

Candidate	CV MAE	CV SD
Ridge alpha=10	9.38	0.51
Ridge alpha=1	9.40	0.53
Boosting leaves=7	10.08	0.55
Random forest leaf=5	10.09	0.66
Random forest leaf=2	10.21	0.68
Boosting leaves=15	10.49	0.64
Median baseline	17.92	0.60

The lowest mean cross-validation MAE belongs to **Ridge alpha=10**. This configuration was selected before test scoring. MAE values represent minutes of absolute error, making them directly interpretable for delivery estimates.

Model	MAE	RMSE	R-squared
Median baseline	17.46	21.18	-0.001
Ridge alpha=10	8.57	11.14	0.723

The selected model has an average absolute error of **8.57 minutes** and reduces MAE by **50.9%** relative to the median baseline. Its RMSE is 11.14 minutes, giving larger mistakes more weight. R-squared is 0.723; it is not a percentage classification accuracy. On this holdout, 66.5% of estimates are within ten minutes of the recorded target.

Fit, error and uncertainty

Training MAE is 9.20 minutes versus 8.57 on the holdout. These values should be read together with validation variation and the small sample size. A favourable random holdout result does not establish stable performance on later operational orders. The ten-minute tolerance rate is descriptive and is not a promised service level.

The excluded preparation duration may explain some remaining variation, but its inclusion would answer a different information-availability question. A practical extension would collect a preparation estimate genuinely available at order time. The estimator must not silently replace that estimate with the later actual duration.

4. Results (continued)

4.2 Error patterns and feature interpretation

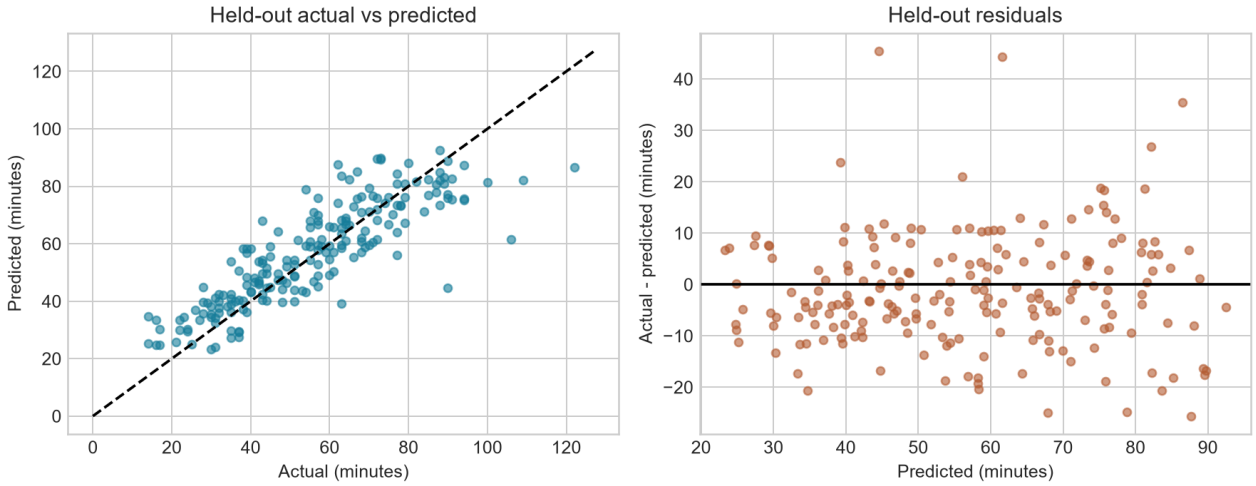


Figure 1. Actual versus predicted duration and residuals on the 200 held-out orders. Points near the diagonal indicate smaller errors; positive residuals indicate underprediction. Difficult long-duration orders can remain poorly estimated even when aggregate metrics are favourable.

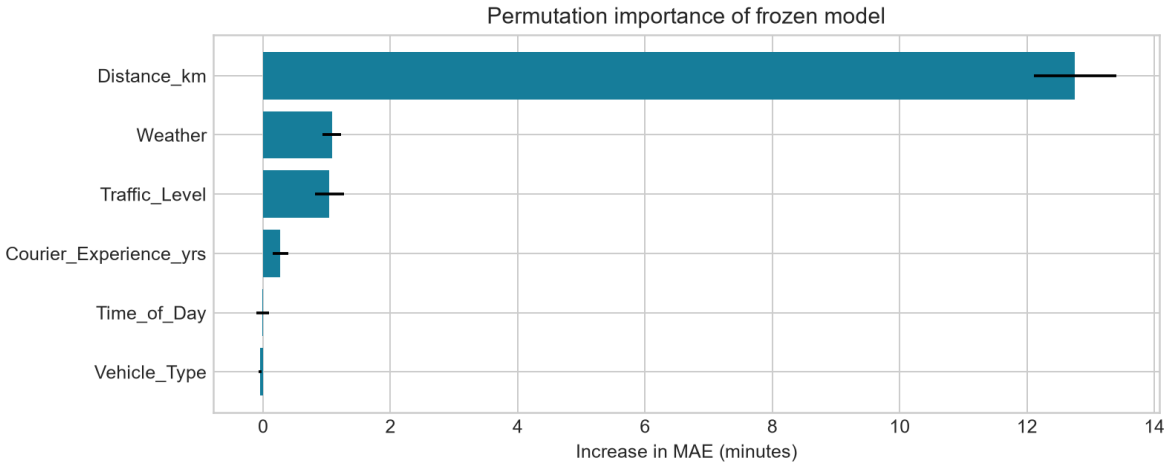


Figure 2. Shuffling **Distance_km** gives the largest mean MAE increase (12.76 minutes). Error bars show variation across ten shuffles. The ranking describes the frozen model and does not establish causal influence. Test-based explanation was not used for additional tuning. [6]

Traffic group	Test orders	MAE (minutes)
High	38	11.26
Low	72	8.40
Medium	79	7.76
Unknown	11	6.22

Group errors include the number of observations so that small groups are not mistaken for precise estimates. Unknown denotes missing traffic information; those records remain in the evaluation.

5. Conclusion

Outcome and KDD learning

The project develops an end-to-end regression model from 1,000 orders. It covers problem identification, data understanding, cleaning, transformation, model comparison, evaluation, interpretation and reuse. The selected Ridge alpha=10 configuration achieves a held-out MAE of 8.57 minutes, RMSE of 11.14 minutes and R-squared of 0.723.

The measured improvement over a constant median indicates that the supplied inputs contain useful predictive information in this sample. It does not demonstrate a reduction in actual delivery delays. The main methodological lesson is to define what information is available before choosing an algorithm, and to keep all learned preprocessing inside the training process.

Reusable model demonstration

The complete pipeline is saved and reloaded successfully. For the first held-out example, it predicts 37.62 minutes, compared with 32 recorded minutes. The inference script reads a CSV containing only the six predictors, validates the schema and writes predictions. Reloaded output agrees with the original model output. This demonstrates local reuse rather than a production deployment.

Limitations

The dataset is small and its operational provenance is not fully documented. There are no dates, restaurant identifiers or courier identifiers for stronger temporal or entity-level validation. Weather and traffic are supplied conditions whose collection timing is unspecified. Menu complexity, kitchen workload and route details are absent, and the estimates do not include calibrated uncertainty intervals.

The missing predictor values are handled consistently, but imputation does not recover the lost information. Random order-level splitting assumes sufficiently independent and comparable observations. Performance could deteriorate under different cities, distances, weather patterns or operating practices. Feature importance is descriptive rather than causal.

Future work

Collect timestamped records and distinguish order-time estimates from later actual measurements. Add a known-at-order kitchen preparation estimate, validate on later dates and unseen entities, compare against a simple distance-and-traffic operational rule, and evaluate calibrated prediction intervals. A monitored deployment should record errors and changes in inputs, with human review of unrealistic outputs.

Delivery estimates should support transparent customer communication. They should not be treated as rigid deadlines that encourage unsafe courier behaviour. The submitted notebook and report provide a reproducible foundation for studying the problem and extending it with better data.

6. References

[1] Den_Kuznetz. Food Delivery Time Prediction. Kaggle, version 1. Dataset metadata lists Apache 2.0.
https://www.kaggle.com/datasets/denkuznetz/food-delivery-time-prediction

[2] Garg, A., Ayaan, M., Parekh, S., and Udandarao, V. (2025). Food Delivery Time Prediction in Indian Cities Using Machine Learning Models. arXiv preprint 2503.15177.
https://arxiv.org/abs/2503.15177

[3] Wen, H., et al. (2023). A Survey on Service Route and Time Prediction in Instant Delivery: Taxonomy, Progress, and Prospects. arXiv preprint 2309.01194.
https://arxiv.org/abs/2309.01194

[4] scikit-learn. Pipelines and composite estimators. Official documentation.
https://scikit-learn.org/stable/modules/compose.html

[5] scikit-learn. Cross-validation: evaluating estimator performance. Official documentation.
https://scikit-learn.org/stable/modules/cross_validation.html

[6] scikit-learn. Permutation feature importance. Official documentation.
https://scikit-learn.org/stable/modules/permutation_importance.html

Sources accessed 11 October 2026. Research discussion uses public abstracts. Exact dependency versions and the dataset checksum are included with the code.

Submission contents

Artifact	Purpose
Food_Delivery_Time_KDD.ipynb	Complete executed implementation and KDD narrative
data/	Dataset, attribution, license, metadata, schema and sample
results/	CV scores, test metrics, predictions and charts
models/food_delivery.joblib	Saved preprocessing and regressor
predict.py / run_notebook.py	Prediction demo and complete notebook rerun
README.md / requirements.txt	Problem, methodology, results and setup
report/	This six-chapter project report